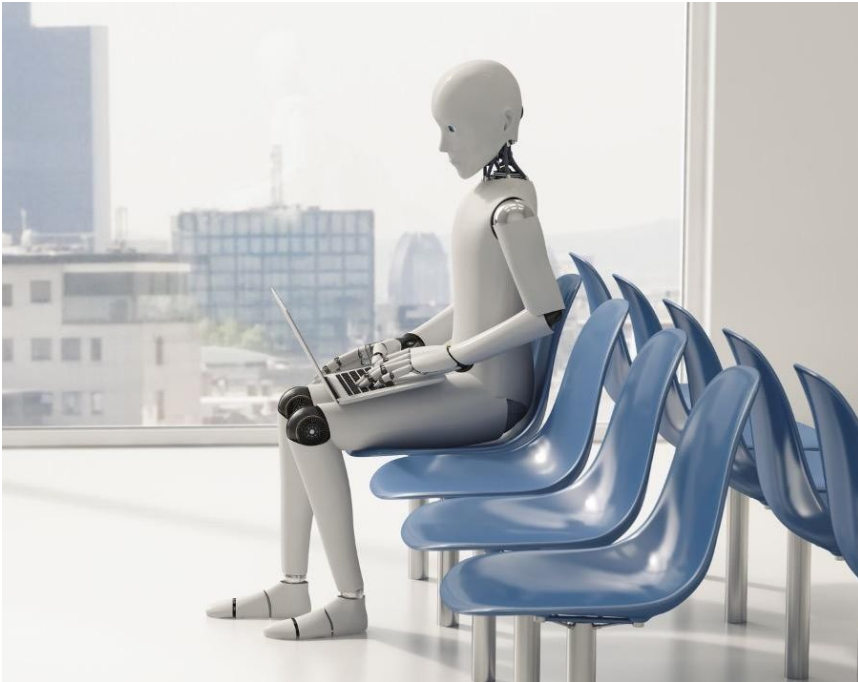




Course: **Building with Claude**

Program Introduction

This 19-hour application engineering program is for technical participants who build with the Claude API – software developers, application developers, data engineers and solution architects. It covers secure API and Python SDK use, reliable prompt design, validated structured (JSON) outputs, tool use and function-style integration with business systems, retrieval-grounded (RAG) responses over enterprise documents, and output evaluation. Hands-on labs and case studies are drawn from finance, retail and telecom contexts, culminating in an applied mini-project that combines prompting, structured output, tool use and validation. Completion of the Working with Claude program or equivalent awareness of Claude is assumed.

Program Objectives

This program is designed to:

- Enable technical participants to use the Claude API and Python SDK securely.
- Develop reliable prompting and validated structured (JSON) outputs.
- Build tool use and function-style integration with business systems.
- Build retrieval-grounded (RAG) responses over enterprise documents.
- Establish output evaluation for accuracy, relevance and safety.

Indicative Design and Content Coverage

Course: Building with Claude

Total Duration: 19 hours

Program at a Glance

Program	Focus	Who it is for	Duration
Building with Claude	Secure API use, structured outputs, tool use, retrieval-grounded responses and evaluation.	Technical participants only: software developers, application developers, data engineers and solution architects building with the Claude API.	19 hours

Key Outcomes

By the end of this program, participants will be able to:

Building with Claude

- Use the Claude API and Python SDK securely, with appropriate key and error handling.
- Design reliable prompts and produce validated structured (JSON) outputs.
- Implement tool use and function-style integration with business systems.
- Build retrieval-grounded (RAG) responses over enterprise documents.
- Evaluate outputs for accuracy, relevance and safety.

Pre-requisites

Building with Claude

- Completion of the Working with Claude program or equivalent awareness of Claude.
- Hands-on Python experience for API-based work.
- Familiarity with REST APIs, JSON and basic application workflows.

Detailed Design

Module	Topics & Detailed Technical Coverage	Hands-on / Case Study	AI Tools / Platforms Used	Duration
Building with Claude · Technical participants only · 19 hours				
API Setup and Secure Integration	Anthropic API access, SDK setup, key and secret handling, request/response structure, error handling, and cost and usage awareness.	Build a Python script that calls Claude securely using environment-managed credentials.	Anthropic Python SDK; environment management	2 hours
Prompt Engineering for Applications	System prompts, instruction hierarchy, few-shot prompting, response formatting, long-context handling, and reducing unsupported output.	Build prompts for a finance credit-policy explanation assistant.	Claude API; prompt patterns	2 hours
Structured Outputs and Validation	JSON generation, schema-driven prompting, parsing, validation, retries, and downstream integration.	Generate validated JSON for a retail product-enrichment task.	Claude API; JSON Schema; Pydantic-style validation	2.5 hours
Conversation and Context Management	Message formatting, conversation memory, summarisation memory, and token-aware design.	Build a telecom complaint summariser with a next-best-action output.	Claude API; long-context techniques	2 hours
Tool Use and Function Integration	Tool-use patterns; calculator, search and database examples; business API integration.	Design tool calls for invoice validation and vendor lookup.	Claude tool use; OpenAPI integration	2.5 hours
Retrieval-Grounded Responses (RAG)	Chunking, embeddings, vector stores, grounding, answer synthesis, and retrieval checks.	Build a finance SOP assistant grounded in source documents.	Claude API; vector store; embeddings	3 hours
Evaluation and Output Quality	Faithfulness, relevance, safety and tool correctness; prompt versioning; human feedback.	Evaluate a RAG assistant for relevance and faithfulness.	Evaluation rubrics; logging	2 hours
Applied Mini-Project	End-to-end assistant design combining prompts, structured output, tool use and validation.	Build a telecom support triage assistant.	Claude API	2 hours
Exit Test	Scenario assessment of application-level skills: prompt design, structured outputs, tool use, retrieval grounding and output evaluation; architecture critique and remediation.	Complete an exit test by designing and partly implementing a small Claude-API feature (structured output, tool call or RAG) for a given enterprise scenario.	Scenario-based exit test	1 hour

Learning Pedagogy

The pedagogic model is focused on experiential learning (in-person and remote virtual learning) mode. Expert mentors work with learners through the program. Learning is in an environment that combines the convenience of anytime access with the intensity of mentoring.

The model combines the following elements:

1. Instructor-led Live Connects: These work on a fixed schedule with recorded versions available to people who missed them.

- Sessions that provide context.
- Sessions that demonstrate the usage of tools or technologies.
- Sessions with expert-led demonstrations that provide step-by-step guidance on critical tasks.
- Sessions that explain best practices.
- Sessions that explain common pitfalls and issues.
- Sessions that discuss success stories, case studies and real-world scenarios that provide insight into the practical challenges and solutions.

2. Reference Learning Material: Curated decks, hands-on labs, sample code, evaluation rubrics and reading lists for self-paced reinforcement between live sessions.

About StackRoute and RPS Consulting



StackRoute and RPS Consulting operate as NIIT's enterprise capability transformation arm, combining advisory, role-based tech skilling, OEM-certified training, assessment, and scaled delivery into one integrated outcome-led platform. The combined entity builds four core capabilities for enterprises: modern engineering (Full-stack, DevOps, SRE, and platform ownership enabling end-to-end engineering excellence), data and AI (data-to-decision systems and production-grade AI deployment), architecture and leadership (developing architects, technology leaders, and transformation owners with system-level thinking), and execution excellence at scale (strengthening project and program management through certifications and delivery governance). The portfolio spans the full talent lifecycle, from campus hiring through senior leadership transformation, organized by business role and career stage, and measured by workforce outcomes rather than training hours.